Nathaniel J. Himmel, PhD

Assistant Professor of Biological Sciences

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POSITIONS & APPOINTMENTS

Assistant Professor 2026 –
New Jersey Institute of Technology, NJ, USA
Federated Department of Biological Sciences (NJIT & Rutgers—Newark)
Mathematical and Computational Biosciences Collective (NJIT)

Premier Assistant/Postdoctoral Fellow 2022 – 2026
Université de Lausanne, Switzerland
Centre Intégratif de Génomique (PI: Richard Benton, PhD FRS)

Research Technician 2013 – 2015
Emory University, GA, USA
Departments of Medicine & Physiology (PI: Mitsi A. Blount, PhD)

EDUCATION

PhD **Neuroscience**, Georgia State University, GA, USA 2022
Neuroscience Institute (PI: Daniel N. Cox, PhD)
Dissertation: *How the Fly Youth Chill: The Molecular Biology, Ecology, and Evolution of Cold Nociception in Drosophila and Subsequent Studies of the Evolution of TRP Channels*

BS **Biology**, University of Florida, FL, USA 2013

AA **Biological Sciences**, Santa Fe College, FL, USA 2011

PUBLICATIONS

* authors contributed equally/co-first authors; co-corresponding authors; [1] undergrad mentee; [2] grad mentee

18. **Himmel NJ**, Moi D, and Benton R. Remote homolog detection places insect chemoreceptors in a cryptic protein superfamily spanning the tree of life. *Current Biology.* 2023. 10.1016/j.cub.2023.10.008

17. Benton R* and **Himmel NJ***. Structural screens identify candidate human homologs of insect chemoreceptors and cryptic *Drosophila* gustatory receptor-like proteins. *eLife*. 2023. 10.7554/eLife.85537

16. **Himmel NJ**, Sakurai A, Letcher JM, Patel AA, Bhattacharjee S, Benson MN[1,2], Gray TR[1], Cymbalyuk GS, and Cox DN. Chloride-dependent mechanisms of multimodal sensory discrimination and nociceptive sensitization in *Drosophila*. *eLife*. 2023. 10.7554/eLife.76863

15. **Himmel NJ** and Benton R. Sweet sensors support stressed cell survival. *PLoS Biology*. 2022. 10.1371/journal.pbio.3001705

14. Patel AA, Sakurai A, **Himmel NJ**, and Cox DN. Modality specific roles for metabotropic GABAergic signaling and calcium induced calcium release mechanisms in regulating cold nociception. *Frontiers in Molecular Neuroscience*. 2022. 10.3389/fnmol.2022.942548

13. **Himmel NJ***, Sakurai A*, Donaldson KJ*, and Cox DN. Protocols for measuring cold-evoked neural activity and cold tolerance in *Drosophila* larvae following fictive cold acclimation. *STAR Protocols*. 2022. 10.1016/j.xpro.2022.101510

12. **Himmel NJ**, Letcher JM, Sakurai A, Benson MN[1,2], Gray TR[1], Donaldson KJ, and Cox DN. Identification of a neural basis for cold acclimation in *Drosophila* larvae. *iScience*. 2021. 10.1016/j.isci.2021.102657

11. **Himmel NJ**, Rogers RT, Redd SK, Wang Y[1], and Blount MA. Purinergic signaling is enhanced in the absence of UT-A1 and UT-A3. *Physiological Reports*. 2021. 10.14814/phy2.14636

10. **Himmel NJ**, Gray TR[1], and Cox DN. Phylogenetics identifies two eumetazoan TRPM clades and an eighth TRP family, TRP Soromelastatin (TRPS). *Molecular Biology and Evolution*. 2020. 10.1093/molbev/msaa065

9. **Himmel NJ** and Cox DN. Transient Receptor Potential Channels: Current Perspectives on Evolution, Structure, Function, and Nomenclature. *Proceedings of the Royal Society B*. 2020. 10.1098/rspb.2020.1309

8. **Himmel NJ***, Letcher JM*, and Cox DN. Dissecting the molecular and neural circuit bases of behavior as an introduction to discovery-driven research: a report on a Course-Based Undergraduate Research Experience. *Journal of Undergraduate Neuroscience Education*. 2020. PMC8040840

7. **Himmel NJ**, Letcher JM, Sakurai A, Gray TR[1], Benson MN[1,2], and Cox DN. *Drosophila* menthol sensitivity and the Precambrian origins of TRP-dependent chemosensation. *Philosophical Transactions of the Royal Society B*. 2019. 10.1098/rstb.2019.0369

6. Lopez-Bellido R, **Himmel NJ**, Gutstein HB, Cox DN, and Galko MJ. An assay for chemical nociception in *Drosophila* larvae. *Philosophical Transactions of the Royal Society B*. 2019. 10.1098/rstb.2019.0282

5. **Himmel NJ**, Rodriguez DA[1], Wang Y[1], Sun MA[1], and Blount MA. Chronic lithium treatment induces novel patterns of pendrin localization and expression. *American Journal of Physiology – Renal Physiology*. 2018. 10.1152/ajprenal.00065.2018

4. **Himmel NJ***, Patel AA*, and Cox DN. Invertebrate Nociception. Originally printed in *The Oxford Research Encyclopedia of Neuroscience*, 2017. Reprinted in *The Oxford Research Encyclopedia of Sensory Systems*, 2022. 10.1093/acrefore/9780190264086.013.166

3. **Himmel NJ** and Cox DN. Sensing the Cold: TRP channels in thermal nociception. *Channels*. 2017. 10.1080/19336950.2017.1336401

2. Turner HN*, Armengol K*, Patel AA, **Himmel NJ**, Sullivan L, Iyer SC, Battacharya S, Iyer EPR, Landry C, Galko MJ, and Cox DN. The TRP channels Pkd2, NompC, and Trpm mediate unique aversive behaviors to noxious cold in *Drosophila*. *Current Biology*. 2016. 10.1016/j.cub.2016.09.038

1. Sim JH, **Himmel NJ**, Redd SK, Pulous FE, Rogers RT, Black LN, Hong SM, von Bergen TN, and Blount MA. Absence of PKC-alpha attenuates lithium-induced nephrogenic diabetes insipidus. *PLoS One*. 2014. 10.1371/journal.pone.0101753

FUNDING & AWARDS

Grants & Funding:

Human Frontier Science Program Long-Term Fellowship (LT-0003/2022-L) 2022–2025
International Human Frontier Science Program Organization
"Evolutionary, expression, and functional characterization of ancient putative chemosensors"

Ruth L. Kirschstein National Research Service Award (F31NS117087) 2020–2022
National Institute of Neurological Disease and Stroke
"Functional roles of chloride homeostasis and chloride ion channels in thermosensory nociception"

Kenneth W. and Georganne F. Honeycutt Fellowship 2017–2020
Georgia State University, Neuroscience Institute

Brains & Behavior Fellowship 2017–2020
Georgia State University, Brains & Behavior Program
Honorary Fellow, 2020-2021

Awards:

Best Poster Prize, *University of Lausanne, Génopode Conference* 2025
Best Poster Prize, *University of Lausanne, Génopode Conference* 2024
Writing Across the Curriculum Pedagogy Award, *Georgia State University* 2021
Neuroscience Institute Outstanding Doctoral Scholar, *Georgia State University* 2020
Neuroscience Institute Outstanding Graduate Student Mentor, *Georgia State University* 2019
Meritorious Research Award, *American Physiological Society* 2016
Meritorious Research Award, *American Physiological Society* 2015

MENTORING, SERVICE, OUTREACH

Mentees:

date	name	location	stage mentored	current position/affiliation
2024	Leana Keel	UNIL	Masters	Masters student, UNIL/Cambridge
2018-2021	Maggie Benson	GSU	Undergrad/Masters	MD/PhD candidate, U Kansas
2017-2020	Thomas Gray	GSU	Undergrad	PhD candidate, Brandeis
2015	Matthew Borchart	Emory U	Undergrad	Physician, U Arizona
2014-2015	Michael Sun	Emory U	Undergrad	Project Manager, GVC
2014-2015	Daniel Rodriguez	Emory U	Undergrad	Equities Trader, Citadel Securities
2014-2015	Yirong Wang	Emory U	Undergrad	Scientific Designer, Promega
2014	John Ward	Emory U	Postbacc	Physician
2013-2015	Nikki A. Mehran	Emory U	Undergrad	Physician, Icahn SOM
2013-2015	Grace Swaim	Emory U	Undergrad	Postdoc, Yale

Editorial service:

Preprint Editorial Team Leader (Physio. & Neuro.), *Proceedings of the Royal Society B* 2019–2023

***Ad hoc* peer review:**

American Journal of Physiology- Regulatory, Integrative and Comparative Physiology | *Biochemical and Biophysical Research Communications* | *The Biological Bulletin* | *Biology Open* | *BMC Genomics* | *Current*

Opinion in Insect Science | *eLife* | *Fish Physiology and Biochemistry* | *Frontiers in Ecology and Evolution* | *Genes, Brain and Behavior* | *Genome Biology and Evolution* | *Journal of Economic Entomology* | *Proceedings of the National Academy of Sciences* | *Science*

Grant review:

Agence Nationale de la Recherche (France)

Media coverage:

Weiner S. "Do insects enjoy sex?" *Gizmodo*. 2017. https://gizmodo.com/doinsects-enjoy-sex-1796376553.

TEACHING

Courses:

Course-Based Undergraduate Research Experience (PERS2002)
Co-designers: Daniel N. Cox & Jamin M. Letcher

Teaching assistantships:

Advanced Molecular, Cellular and Developmental Biology (BSc3)	2023, 2024, 2025
Biostatistics (BSc2)	2023
Neuroscience Laboratory (NEUR3010/4000)	2016, 2018
Course-Based Undergraduate Research Experience (PERS2002)	2017

Other teaching:

Writing Advisor - Writing Across the Curriculum Program 2018–2021
GSU, Center for Excellence in Teaching and Learning

Guest lectures:

GSU's Neuroscience School: *Presenting Data in Different Forms*	2019
GSU's Neuroscience School: *Neurological Diseases*	2016
Atlanta Brain Bee Prep Course: *Senses, Perception, and Movement*	2015
Emory University SUPERR Program: *Techniques in Renal Physiology*	2015

PRESENTATIONS, POSTERS, ABSTRACTS

Invited seminars & symposia:

6. University of Fribourg, Department of Biology. Invited seminar. Fribourg, Switzerland. 2024.

5. Georgia State University, Neuroscience Institute. Invited seminar. Virtual. 2022.

4. North Carlina State University, College of Veterinary Medicine, Fungal Systems Biology Laboratory. Invited seminar. Raleigh, NC, USA. 2021.

3. Monthly Maggot Meeting. Invited seminar. Virtual. 2021.

2. Theo Murphy International Scientific Meeting on the evolution of mechanisms and behaviour important for pain, hosted by The Royal Society. Invited participant. Chicheley, Buckinghamshire, UK. 2019

1. NIDDK/KUH Researcher Panel, Summer Student Symposium. Invited panelist. Atlanta, GA, USA. 2015.

Conference talks:

7. **Himmel NJ**, Keel L, Fournes F, Collier J, and Benton R. From the Depths of the Twilight Zone: expression & functional analyses of an Odorant & Gustatory receptor homolog conserved in flies, humans and beyond. October 2025. *Neurobiology of Drosophila, Cold Spring Harbor, NY, USA.*

6. **Himmel NJ**, Keel L, Fournes F, Collier J, and Benton R. From the Depths of the Twilight Zone: expression & functional analyses of an Odorant & Gustatory receptor homolog conserved in flies, humans and beyond. September 2025. *19th biennial European Society for Insect Taste and Olfaction, Villasimius, Italy.*

5. **Himmel NJ**, Keel L, Fournes F, Collier J, and Benton R. Twilight zone homologs of insect Odorant receptors are evolutionarily constrained, sperm-expressed proteins conserved in humans, flies, and beyond. July 2024. *3rd Joint Congress on Evolutionary Biology, Montreal, QC, Canada.*

4. **Himmel NJ** and Benton R. Evolutionary, expression, and functional analyses of a cryptic Or/Gr homolog conserved in flies, humans, and beyond. June 2024. *19th International Symposium on Olfaction and Taste, Reykjavík, Iceland.*

3. **Himmel NJ**, Moi D, Benton R. The enigmatic origins of insect Gustatory and Odorant receptors: a homologous protein superfamily spanning eukaryotes, archaea, and bacteria. June 2023. *Symposium on the emergence of order across biological scales, Lausanne, Switzerland.*

2. **Himmel NJ**, Letcher JM, Sakurai A, Gray TR, Benson MN, Cox DN. Menthol elicits *Trpm*- and *TrpA1*-dependent rolling in *Drosophila* larvae, suggesting Precambrian origins for TRP-dependent menthol sensing. October 2019. *Neurobiology of Drosophila, Cold Spring Harbor, NY, USA.*

1. **Himmel NJ,** Rodriguez DA, Blount MA. Chronic lithium treatment induces β-intercalated cell expression in the renal inner medulla. March 2015. *Experimental Biology, Boston, MA, USA.*

Posters (only first/last-authored) & published abstracts (all):

32. **Himmel NJ**, Keel L, Fournes F, Collier J, and Benton R. From the Depths of the Twilight Zone: an Or/Gr homolog conserved in flies, humans and beyond. September 2025. *Annual Génepode Retreat, Leysin, Switzerland.*

31. **Himmel NJ**, Keel L, Fournes F, Collier J, and Benton R. A testis-expressed homolog of insect Gustatory and Odorant receptors. September 2024. *Annual Génepode Retreat, Leysin, Switzerland.*

30. **Himmel NJ**, Keel L, Fournes F, Collier J, and Benton R. A testis-expressed homolog of insect Gustatory and Odorant receptors, July 2024. *Swiss Scientific Advisory Committee meeting, Lausanne, Switzerland.*

29. **Himmel NJ**, Keel L, Fournes F, Collier J, and Benton R. A testis-expressed homolog of insect Gustatory and Odorant receptors. June 2024. *Annual Swiss Drosophila Meeting, Lausanne, Switzerland.*

28. **Himmel NJ** and Benton R. Evolutionary, expression, and functional analyses of a cryptic Or/Gr homolog conserved in flies, humans, and beyond. in press, *Chemical* Senses, June 2024.

27. **Himmel NJ**, Moi D, and Benton R. The mysterious origins of insect Odorant receptors; a case study for remote homolog detection in the protein folding age. December 2023. *22nd Annual Human Frontier Science Program Awardees Meeting, Cape Town, South Africa.*

26. **Himmel NJ**, Moi D, and Benton R. The enigmatic origins of *Drosophila* Gustatory and Odorant receptors. June 2023. *Annual Génepode Retreat, Leysin, Switzerland.*

25. **Himmel NJ**, Moi D, and Benton R. The enigmatic origins of *Drosophila* Gustatory and Odorant receptors. June 2023. *Annual Swiss Drosophila Meeting, Fribourg, Switzerland.*

24. **Himmel NJ** and Benton R. Ancient putative chemosensors in the zoosporic fungus *Spizellomyces punctatus*. June 2022. *Annual Génepode Retreat, Les Diablerets, Switzerland.*

23. **Himmel NJ** and Benton R. Ancient putative chemosensors in the zoosporic fungus *Spizellomyces punctatus*. June 2022. *Symposium on interactions in biology, University of Lausanne, Lausanne, Switzerland.*

22. **Himmel NJ,** Sakurai A, Letcher JM, Patel AA, Benson MN, Bhattacharjee S, Gray TR, Cymbalyuk GS, and Cox DN. Chloride-dependent mechanisms of multimodal sensory discrimination: roles for *subdued*, *koozie*, and excitatory chloride physiology in *Drosophila* cold nociception. October 2021. *Neurobiology of Drosophila Meeting, Cold Spring Harbor Lab, virtual.*

21. **Himmel NJ**, Letcher JM, Sakurai A, Benson MN, Gray TR, Donaldson KJ, and Cox DN. How The Fly Youth Chill. March 2021. *Annual GSU Brains & Behavior Retreat, virtual.*

20. **Himmel NJ**, Letcher JM, Sakurai A, Benson MN, Gray TR, Donaldson KJ, and Cox DN. How The Fly Youth Chill: The evolution of cold nociception in drosophilid larvae and identification of a neural basis for cold acclimation. March 2021. *Animal Behavior Conference*, *hosted by Indiana University—Bloomington, virtual.*

19. **Himmel NJ**, Sakurai A, Letcher JM, Benson MA, Gray TR, and Cox DN. Excitatory chloride physiology discriminately encodes noxious cold in a multimodal sensory neuron. January 2021. *Annual meeting of the Society for Neuroscience, virtual*.

18. Maksymchuk N, Sakurai A, Patel AA, **Himmel NJ**, Cox DN, Cymbalyuk G. Mechanisms of cold temperatures rate coding by *Drosophila* CIII neurons. October 2019. *Annual meeting of the Society for Neuroscience, San Diego, CA, USA*. Abstract online.

17. Letcher JM, **Himmel NJ**, Sakurai A, Holgiun-Lopez M, Cox DN. TrpA1 mediates cold nociception in *Drosophila melanogaster* larvae. October 2019. *Neurobiology of Drosophila Meeting, Cold Spring Harbor Lab, Cold Spring Harbor, NY, USA.*

16. Maksymchuk N, Sakurai A, Patel AA, **Himmel NJ**, Cox DN, Cymbalyuk G. Role of TRP channels in temperature rate coding by *Drosophila* noxious cold sensitive neurons. *BMC Neuroscience*, 20:56, July 2019. doi: 10.1186/s12868-019-0538-0

15. **Himmel NJ**, Letcher JM, Gray TR, Benson MN, Cox DN. TRP-dependent chemical sensing: the Precambrian hypothesis. May 2019. *Annual GSU Brains & Behavior Retreat, Atlanta, GA, USA.*

14. **Himmel NJ**, Letcher JM, Gray TR, Benson MN, Cox DN. Menthol elicits a *Trpm*- and *TrpA1*-dependent nocifensive response in *Drosophila melanogaster* larvae. February 2019. *Theo Murphy International Scientific Meeting on the evolution of mechanisms and behaviour important for pain, hosted by The Royal Society, Chicheley, Buckinghamshire, UK*.

13. Patel AA, **Himmel NJ**, Cox DN. Calcium induced calcium release mechanisms in cold nociception. February 2019. *Theo Murphy International Scientific Meeting on the evolution of mechanisms and behaviour important for pain, hosted by The Royal Society, Chicheley, Buckinghamshire, UK*.

12. Patel AA, **Himmel NJ**, Yang JJ, Cox DN. Cellular and behavioral requirements for calcium release mechanisms in cold nociception. October 2018. *Annual meeting of the Society for Neuroscience, San Diego, CA, USA.* Abstract online.

11. Maksymchuk N, Patel AA, **Himmel NJ**, Cox DN, Cymbalyuk G. Modeling of TRP channel mediated noxious cold sensation in *Drosophila* sensory neurons. *BMC Neuroscience*, 19(Suppl 2):64, July 2018. doi: 10.1186/s12868-018-0452-x

10. **Himmel NJ**, Letcher JM, Gray TR, Cox DN. The evolution of cold nociception in drosophilid larvae. May 2018. *Annual GSU Brains & Behavior Retreat, Atlanta, GA, USA.*

9. **Himmel NJ**, Gray TR, Cox DN. Anoctamins are required for cold nociception in *Drosophila.* April 2018. *Annual Georgia Collegiate Neuroscience Symposium*, *Athens, GA, USA.*

8. **Himmel NJ**, Gray TR, Cox DN. Calcium-activated chloride channels are required for distinguishing between noxious and innocuous stimuli in multimodal sensory neurons. April 2018. A*nnual meeting of the Genetics Society of America, Philadelphia, PA, USA.* Abstract online.

7. Maksymchuk N, Patel AA, **Himmel NJ**, Cox DN, Cymbalyuk G. Modeling cellular noxious cold sensation in *Drosophila* sensory neurons. October 2017A*annual meeting of the Society for Neuroscience, Washington DC, USA*. Abstract online.

6. Kronk TA, **Himmel NJ**, Mehran NA, Blount MA. Lithium administration can attenuate the progression of polycystic kidney disease. *The FASEB Journal*, 31(Suppl 1):1032.4, April 2017. doi: 10.1096/fasebj.31.1_supplement.1032.4

5. Patel AA, Moon D, **Himmel NJ**, Cox DN. Cellular and molecular dissection of noxious cold nociception in *Drosophila*. March 2017. *58th Annual Drosophila Research Conference, San Diego, CA, USA.* Abstract.

4. Patel AA, Turner HN, Armengol K, **Himmel NJ**, Galko MJ, Cox DN. Cellular and molecular dissection of noxious cold nociception in *Drosophila*. July 2016. *The Allied Genetics Conference, Orlando, FL, USA*. Abstract online.

3. **Himmel NJ**, Rodriguez DA, Wang Y, Sun MA, Blount MA. Chronic lithium treatment induces novel patterns of pendrin localization and expression in the kidney. *The FASEB Journal*, 30(Suppl1):968.14, April 2016. doi: 10.1096/fasebj.30.1_supplement.968.14

2. **Himmel NJ**, Rogers RT**,** Redd SK, Blount MA. Purinergic signaling is enhanced in the absence of UT-A1 and UT-A3. October 2014. *Emory Department of Medicine Research Day, Atlanta, GA, USA.*

1. Rogers RT, **Himmel NJ,** Redd SK, Blount MA. Purinergic signaling is enhanced in the absence of UT-A1 and UT-A3. *The FASEB Journal*, 38(Suppl 1): 969.17, April 2014. doi: 10.1096/fasebj.28.1_supplement.1137.10

Institutional oral presentations:

7. From the Depths of the Twilight Zone: an Odorant & Gustatory receptor homolog conserved in flies, humans and beyond. January 2026. *Center for Integrative Genomics Progress Report Seminar*, *Lausanne, Switzerland.*

6. Chasing cryptic protein homologues through the twilight zone of molecular evolution. March 2023. *Center for Integrative Genomics Progress Report Seminar*, *Lausanne, Switzerland.*

5. How The Fly Youth Chill: The protective role of cold nociception in *Drosophila* larvae. June 2021. *Neuroscience Institute Breakfast and Lecture Series, Atlanta, GA, USA.*

4. Insects and vertebrates are the odd ones out: unexpected findings in the evolution of TRP channels. September 2019. *Neuroscience Institute Breakfast and Lecture Series, Atlanta, GA, USA.*

3. Multimodality in *Drosophila melanogaster* sensory neurons. February 2018. *Neurogenomics Forum, Atlanta, GA, USA.*

2. Calcium-activated chloride channels play a role in the function of multimodal sensory neurons. August 2017. *Neuroscience Institute Breakfast and Lecture Series, Atlanta, GA, USA.*

1. Chronic lithium treatment induces renal β-intercalated cell expression in the inner medulla. October 2014. *Emory Transport Signaling Workers Union Seminar, Atlanta, GA, USA.*